

Official Agricultural Extension Advisory: False Smut in Rice (Paddy)

Causal Pathogen: Ustilaginoidea virens (teleomorph Villosiclava virens)

Issuing Source: Directorate of Plant Protection, Quarantine & Storage (DPPQS) & ICAR-IIRR | Date: 2024-04-05

1. Identification and Morphological Symptoms

Symptoms become visible only after flowering and grain development. Individual spikelets on the panicle are transformed into large, velvety green or orange-yellow spore balls, usually more than twice the diameter of normal grains. Initially, the spore balls are small, yellow, and covered with a smooth membrane. Later, the membrane bursts, turning the ball dark olive-green or greenish-black, releasing powdery chlamydospores that contaminate neighboring grains.

2. Environmental Epidemiology & Causes

Favored by high relative humidity (>90%) and cloudy, rainy weather during the flowering/heading stage, with moderate temperatures of 25-30 deg C. Excessive nitrogen application at boot stage significantly increases spikelet susceptibility.

3. Cultural Management and Field Prevention

Prevention: Use certified clean seeds free from smut balls. Avoid excessive nitrogen top dressing after panicle initiation. Adjust transplanting dates to avoid flowering during continuous cloudy/rainy spells.

Cultural Practices: Manually collect and safely burn false smut balls wrapped in polythene bags to prevent spore dissemination. Deep summer plowing to bury resting sclerotia below the topsoil.

4. Organic and Biological Management

Seed treatment with *Trichoderma asperellum* @ 10 g/kg seed. Foliar spray of cow urine formulation (10%) mixed with neem oil (2%) at early boot leaf stage. Application of bio-agent *Bacillus subtilis* @ 5 g/L during boot stage.

5. Verified Chemical Interventions

Preventive spray at boot leaf stage (before panicle emergence): Copper Hydroxide 77% WP @ 2.0 g/L or Propiconazole 25% EC @ 1.0 mL/L or Trifloxystrobin 25% + Tebuconazole 50% WG @ 0.4 g/L. Spraying after smut balls appear is ineffective.

• **Active Ingredient:** Trifloxystrobin 25% + Tebuconazole 50% WG | **Reference Formulation:** Nativo (standard registered formulation) | **Registrant:** Bayer CropScience Ltd | **Source:** DPPQS CIBRC Registered Fungicide Gazette (2024-02-20)

• **Active Ingredient:** Copper Hydroxide 77% WP | **Reference Formulation:** Kocide 2000 | **Registrant:** Certis / Kocide LLC | **Source:** ICAR-IIRR Advisory 2024 (2024-03-10)

Notice: Chemical recommendations are derived strictly from central agricultural research advisories. Farmers must consult local district agricultural extension officers for local dosage and label claims.